
Numerical Methods Sample Exam

1. Consider the problem of finding a solution of the algebraic equation $e^x - 6x - 2 = 0$ in the interval $[0, 4]$.
 - (a) Compute one step of Newton's algorithm, starting from the midpoint of the interval.
 - (b) Is the value obtained within the interval $[0, 4]$? Why?
 - (c) Perform one step of the bisection method to find a smaller bracket for the solution.
 - (d) Perform one step of Newton's method starting from the midpoint of the interval (bracket) you found in part (c).
 - (e) Do you expect your new approximation to be within 0.1 of the true solution? Give a reason for your answer.
2. Consider the problem of finding the value of $y(3)$ for the initial value problem

$$\frac{dy}{dt} = \frac{1}{t-1} - y^2, \quad y(2) = 14.$$

- (a) Solve the problem using Ralston's method with a step size of $h = 0.5$.
 - (b) How would you expect the error to change if you were to take a stepsize of $h = 0.25$?
3. Consider the following data

x_i	2.0	1.5	1.0	3.0
$y_i = f(x_i)$	0.40	1.00	1.82	0.03

- (a) Compute the divided differences $f[x_i, \dots, x_j]$ for $0 \leq i \leq j \leq 3$. You may assume $f[x_0, x_1] = -1.200$ and $f[x_1, x_2] = -1.640$.
 - (b) Write down the nested form of the cubic polynomial p_3 interpolating the given data.
 - (c) Estimate the value of y when $x = 2.5$, showing intermediate steps in your computations.

4. Consider the problem of computing to an accuracy of 0.01 the integral

$$I = \int_{0.0}^{1.2} \cos(x^2) dx.$$

- Use the trapezoid rule with $n = 4$ subintervals to estimate the value of I . Work in 4dp.
 - Use the error estimate for the trapezoid rule to estimate the error over the subintervals $[0.0, 0.6]$ and $[0.6, 1.2]$. Do you expect the error to be less than the desired accuracy bound?
 - Explain how to use the adaptive trapezoid method to obtain a better estimate to the integral. (You do not have to perform the computation).
5. The least squares approximation to a function f over the interval $[-1, 1]$ is given by

$$q_n(x) = \sum_{k=0}^n c_k P_k(x)$$

where P_k are the Legendre polynomials. Suppose c_k for $k = 0, \dots, 5$ are given by

k	0	1	2	3	4	5
c_k	0.3086	0.2193	-0.0842	-0.1233	-0.0267	0.0262

- Use the recurrence equation to calculate $P_k(0.7)$ for $k = 0, 1, \dots, 4$.
- Compute $q_4(0.7)$.
- Explain how orthogonality of the P_k gives rise to the following simple formula for the square integral

$$\int_{-1}^1 q_n(x)^2 dx = \sum_{k=0}^n c_k^2 \int_{-1}^1 P_k(x)^2 dx.$$

- Compute the square integral $\int_{-1}^1 q_2(x)^2 dx$.

6. Let

$$A = A^{(0)} = \begin{pmatrix} 5 & 2 \\ 2 & 3 \end{pmatrix}.$$

After applying one step of QR method we obtain

$$A^{(0)} = Q^{(0)} R^{(0)} = \begin{pmatrix} 0.9285 & -0.3714 \\ 0.3714 & 0.9285 \end{pmatrix} \begin{pmatrix} 5.3852 & 2.9711 \\ 0 & 2.0426 \end{pmatrix}; \quad A^{(1)} = R^{(0)} Q^{(0)} = \begin{pmatrix} 6.1036 & 0.7586 \\ 0.7586 & 1.8966 \end{pmatrix}.$$

- Apply one step of the QR method to the matrix $A^{(1)}$.
- Use your answer to part (a) to estimate the eigenvalues of A .